MURTAZA ZAIDI

AI Software Engineer

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EXPERIENCE

Kiresult | AI Software Engineer Dec 2025 – Present

Bonn, Germany (Remote)

- Contributed to KIRA, an AI procurement assistant using LangChain and Pydantic AI, resolving 50–60% of standard analyst queries through tool calling and automated spend analysis.
- Implemented Jev for model and tool routing and RAG relevance filtering to select valid tool combinations and remove irrelevant document results.
- Developed specialized agents for spend analysis, vendor insights, and document retrieval, supporting 4+ procurement workflows.
- Built guardrails and evaluated tool calls, retrieval, and answer quality across 50+ test cases.

BASF | Data Analytics & AI Trainee Mar 2024 – Oct 2025

Berlin, Germany

- Reduced manual patent triage by 40% with a patent-search system using Pydantic AI, FastAPI, LangChain, MongoDB, and structured outputs.
- Designed RAG components and semantic chunking to retrieve relevant patent evidence and generate structured Excel reports.
- Built ETL pipelines to load chemical ontology and knowledge-graph data into relational databases.
- Benchmarked AI models on chemical data using precision, recall, F1, and ROC-AUC.

Udhaar App (Y Combinator Sartup) | Python Engineer Oct 2021 – Sep 2023

Karachi, Pakistan

- Built Django/Celery services for invoicing, ledgers, and payments serving 100K+ concurrent users.
- Automated risk analytics and reporting with Airflow and AWS; improved load times by 30% through PostgreSQL and Redis optimization.
- Reduced fraud by 35% by adding ML/NLP detection to KYC checks and transaction processing.

PROJECTS

Data-Man — Agentic Data Analytics Assistant

Python, LangGraph, LangChain, PostgreSQL, Pandas, TypeSafeAI JEV, RAG, Streamlit

- Engineered an Agentic workflow that coordinates SQL analysis and ETL agents, enabling users to query PostgreSQL data and request data transformations in natural language.
- Implemented schema-aware RAG to retrieve relevant table and column context for SQL generation.
- Integrated TypeSafeAI JEV for intelligent tool-call selection, guardrail enforcement, and evaluation of routing decisions, tool execution, and response quality.

Hasso Plattner Institute | Master's Thesis

- Developing an LLM inference placement layer that routes requests across heterogeneous GPU tiers using output-length prediction and load-aware scheduling to minimize costs while meeting latency SLOs.
- Evaluating tier-aware placement against single-tier and FCFS baselines using vLLM, measuring latency, financial cost, and energy consumption across diverse workloads.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, C++

AI Engineering: LangGraph, LangChain, Pydantic AI, JEV, RAG, Multi-Agent Systems, Guardrails, RAG & LLM Evaluation, vLLM

Machine Learning: PyTorch, Scikit-Learn, Feature Engineering, Neural Networks, Computer Vision

Backend & Data: FastAPI, Pydantic, Celery, Streamlit, PostgreSQL, MongoDB, ChromaDB, Redis, Django

Cloud & DevOps: Docker, CI/CD, Git, Linux

EDUCATION

University of Potsdam | Master's in Data Science GPA: 1.9 (German scale)

NED University | Bachelor's in Computer Systems GPA: 1.5 (German scale)